Unexpectedness by Design: Causal Serendipity Optimization for Implicit Recommender Systems

Final Project

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Note: All code and relevant data sources for this project can be found on GitHub.

1 Introduction

Recommender systems serve as the primary backbone for discovery and consumption across modern digital platforms. Matrix factorization techniques have historically served as the engine for these systems, optimizing for accuracy with respect to the system's ability to predict a user's affinity for unseen items based on their historical interactions. The "Netflix Prize," a competition that spanned 2006–2009, is an example of this early motivation, awarding $1 million to any team that was able to increase recommendation accuracy, measured by root mean squared error, by over 10%. However, Wang et al. argue that because these models train on observational data where users actively self-select the content they consume, they suffer from exposure bias. Through the lens of a causal inference framework, by fundamentally assuming that unobserved interactions are missing completely at random (MCAR), traditional matrix factorization conflates mainstream popularity with unconfounded user preference [Wang et al., 2020]. The result is a feedback loop that traps users in algorithm-driven monocultures, repeatedly surfacing familiar, highly expected, and culturally ubiquitous content. To break this cycle and foster true discovery, systems can instead optimize for *serendipity*, an emerging concept that, while not well-defined, roughly describes the delivery of recommendations that are simultaneously highly relevant and delightfully unexpected [Kotkov et al., 2016].

In this paper, we propose a novel algorithm that re-frames recommendation as a causal inference problem to optimize for serendipity by design. Drawing on Wang et al.'s deconfounding framework and Johnson's logistic matrix factorization (LMF) framework for implicit user feedback, we attempt to isolate a user's pure, unconfounded preference from the platform's overarching popularity bias. Building on this causal foundation, we propose a serendipity score that maximizes relevance while penalizing the probability of natural exposure via a method similar to inverse propensity weighting (IPW). Following the discussion of our results, we propose additional enhancements to our proposed algorithm. Particularly, recognizing that an obscure recommendation slate risks user fatigue, we subsequently outline how Peng et al.'s consumption constraints can be integrated to balance serendipitous discoveries with baseline reliability, formalizing a practically implementable recommender system.

Section 2 details the theoretical foundations of causal recommender systems, implicit feedback modeling, and our formal definition of serendipity. Section 3 reiterates our algorithmic pipeline, and Section 4 presents the results of our algorithm on a sparse, implicit dataset. Finally, Section 5 proposes potential enhancements, including the incorporation of utility in the final recommender system as aforementioned.

2 Motivation from Existing Literature

2.1 Recommender Systems as a Causal Inference Problem

The task of recommender systems is classically framed as the need to accurately predict users' preferences and ratings. Wang et al. argue that recommendations are fundamentally a causal inference question: posed in the context of movie recommendations, what would be a viewer's rating if we forced them to watch a given movie? The issue here is that traditional matrix factorization methods only provide valid causal predictions if users interact with items (in this context, movies) purely at random. In terms of the Neyman-Rubin causal model, every user-item pair has two potential outcomes:

- $Y_{ui}(1)$: The rating user u would give item i if they were exposed to it (e.g., forced to watch a movie).
- $Y_{ui}(0)$: The rating the user would give the item i if they were not exposed to it.
- $a_{ui} \in \{0, 1\}$: The treatment assignment (e.g., whether they actually watched the movie).
- *Note:* When modeling the potential outcomes (or "ratings") noted in the first two bullets, we will use y_{ui} to represent predicted rating.

Recall that in reality, we only observe the factual outcome: $Y_{ui} = Y_{ui}(a_{ui})$. Said differently, if a user hasn't seen a movie, $a_{ui} = 0$, and so $Y_{ui}(1)$ is "missing." To satisfy the MCAR property (missing completely at random), it must hold that the treatment assignment, or exposure, is independent of the potential outcomes:

$$a_{ui} \perp Y_{ui}(1), Y_{ui}(0)$$

If MCAR holds, this implies that users discover and consume items completely by chance. For example, under MCAR, the average observed rating for *Imitation Game* from people who actually watched the movie is a perfectly unbiased estimate of how everyone in the world would rate *Imitation Game*—including those who haven't watched it.

In reality, as Wang et al. note, and as the reader may quickly intuit, users interact with items based on a variety of hidden preferences. For example, a user who loves Korean dramas is far more likely to seek out new Korean dramas and watch them. This implies that the treatment ($a_{ui} = 1$) is dependent on the potential outcome $Y_{ui}(1)$ being high, breaking the MCAR assumption. Wang et al. argue that this violation of MCAR serves as a form of confounding bias: the variables that cause the data to be missing (e.g., a user's genre preferences, location, affinity for a director) serve as unobserved confounders that influence the final rating [Wang et al., 2020].

This poses an issue for traditional matrix factorization (MF) methods, which assume MCAR holds. At a *very high level*, MF represents users as rows and items as columns; corresponding cells contain ratings. We expect this matrix to be extremely sparse, as any given user is unlikely to have watched, for example, the entire Netflix catalog. Traditional MF tries to discover hidden (latent) features about both the users and items:

1. MF breaks the sparse matrix into two smaller, dense matrices: (1) a **user matrix** that represents each user as a vector of features (e.g., how much they like action or romance) and (2) an **item matrix** that represents each movie as a vector of numbers across these hidden features (e.g., how much action or romance a movie has).
2. To predict what user A would rate item B, MF takes the **dot product** of user A's latent vector and item B's latent vector. If their vectors align (e.g., the user loves action and the movie is an action film), the predicted rating is high.
3. The algorithm determines the values for these latent vectors by calculating a prediction for an observed rating, comparing it to the actual rating, measuring the error, and minimizing with respect to a loss function.

In step 3, the standard approach is to minimize MSE, or the squared difference between the true rating y and predicting rating $\hat{y}$, summed only over the user-item pairs that were actually observed:

$$\text{Minimize} \sum_{(u,i)\in\text{Observed}} (y_{ui} - \hat{y}_{ui})^2$$

Aside. The above is a simplified version of MF's objective function. More formally, let the matrix of observed ratings be Y. We represent each user u with a latent preference vector θ_u and each item i with a latent attribute vector β_i. The model predicts a rating via their dot product: $\hat{y}_{ui} = (\theta_u^\top \beta_i)$. Incorporating L2 regularization (Ridge penalty) to prevent overfitting by penalizing excessively large weights, we formally aim to minimize the objective function:

$$\min_{\Theta,B} \sum_{(u,i)\in\text{Observed}} (y_{ui} - \theta_u^\top \beta_i)^2 + \lambda(||\theta_u||^2 + ||\beta_i||^2)$$

which can be solved via stochastic gradient descent, alternating least squares, or similar optimization algorithms.

Immediately apparent in both the simplified and complete formulas is that unobserved items are ignored as part of MF's minimization. By calculating the error only on observed ratings, MF assumes that the observed ratings are an unbiased, random sample of the entire matrix (which includes unobserved ratings). Said differently, it assumes that if it can accurately predict 1% of movies a user rated, the same model can be used to predict the 99% of movies a user didn't rate. As Wang et al. point out, this only provides valid predictions if users randomly watch movies, which we argued earlier is unrealistic due to confounding variables, which include selection bias.

Since users self-select what they watch, the observed data is skewed towards positive ratings. People don't tend to watch and rate movies they know they won't like; if a user knows they hate horror movies, they won't watch (and therefore, won't rate) them. Since traditional MF assumes MCAR, it doesn't view missingness as a signal of dislike, but rather just ignores it altogether. The model thus ends up predicting artificially high ratings for, in our example, horror movies, because it is generalizing from a biased sample that this user generally gives 4s and 5s.

Wang et al. posit a method that directly addresses this violation of MCAR. They fit Poisson factorization to the binary exposure matrix (a_{ui}), effectively modeling the exact mechanism of the missing data. Said differently, it determines probabilities that a user will naturally encounter a movie—probabilities that are affected by the confounding variables we outlined above. The output of this model, $\hat{a}_{ui}$, or the expected probability of exposure, is passed into the final outcome model to serve as a substitute confounder. Rather than predicting the rating via a simple dot product (the traditional MF prediction, $\theta_u^\top \beta_i$), Wang et al. modify the outcome equation to condition on this exposure probability:

$$y_{ui}(a) = \theta_u^\top \beta_i \cdot a + \gamma_u \cdot \hat{a}_{ui} + \epsilon_{ui}$$

Here, γ_u is a user-specific coefficient that absorbs the bias of the exposure mechanism. By absorbing confounding bias into the $\gamma_u \cdot \hat{a}_{ui}$ term, the remaining latent vectors ($\theta_u^\top \beta_i$) successfully isolate the true, unconfounded user preference [Wang et al., 2020].

2.2 Modeling Implicit Feedback

Thus far, Wang et al.'s causal inference framework provide a strong theoretical foundation for serendipity in the context of a recommender system. However, this framework operates under the assumption of known, explicit outcomes. For example, Wang assumes we can observe or predict a true rating y_{ui}. In reality, explicit feedback (e.g., 5-star ratings) are rare; rather, platforms must rely on implicit feedback, such as stream counts, clicks, or watch times. To operationalize our recommender system, we must bridge the gap between implicit behavior and explicit preference.

Fortunately, Johnson's framework for logistic matrix factorization can aid us. Johnson proposes that raw interaction volume (e.g., the number of times a user streams a song), denoted as r_{ui}, can be treated as a measure of confidence, which can stand in for an explicit rating. From the implicit data, Johnson defines a confidence metric:

$$c_{ui} = \alpha r_{ui}$$

where α is a scaling hyperparameter. As r_{ui} increases, the system's confidence that the user genuinely prefers the item increases [Johnson, 2014].

Crucially, Johnson's framework clearly defines our optimization problem. While Wang et al.'s causal framework relies on minimizing mean squared error, which fails on highly sparse implicit datasets, Johnson's logistic loss optimization navigates the sparsity of our data by treating unobserved interactions as low-confidence negatives, rather than zeroes. That is, it utilizes interaction volume as a confidence weight within the objective function, preventing the overwhelming number of unobserved items from collapsing the latent vectors to zero. Johnson proposes the following logistic function:

$$p(l_{ui}) = \sigma(\theta_u^\top \beta_i)$$

where $p(l_{ui})$ is the probability that a user likes an item and σ is the standard logistic sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

To combine this with our causal objective, we inject Wang's substitute confounder directly into the logistic equation, yielding our causally-conditioned logistic prediction:

$$p(l_{ui}) = \sigma(\theta_u^\top \beta_i + \gamma_u \hat{a}_{ui})$$

By employing Johnson's confidence-weighted logistic loss function to train this modified equation, the parameter γ_u successfully absorbs the exposure bias ($\hat{a}_{ui}$) within the logistic curve, allowing the algorithm to accurately learn the true, unconfounded latent vectors (θ_u and β_i) via alternating gradient ascent.

Aside. Because the objective function maximizes the log-posterior of a probability rather than minimizing an error, Johnson optimizes via gradient ascent. Additionally, since the dot product of two unknown latent matrices ($\theta_u^\top \beta_i$) is non-convex, we alternate the parameter updates, fixing item vectors to update user vectors, and vice versa, to reduce the optimization to a series of convex logistic regressions.

2.3 Defining Serendipity

In recent literature, the concept of *serendipity* has emerged with a variety of definitions, succinctly summarized by Kotkov. For music recommendations, serendipity can loosely be described as a user's desire to feel as if they are exploring new depths as they listen to new music. In the context of recommender systems, we propose that a serendipitous recommendation is one that is both surprising and satisfying. However, in the traditional recommender system framework based on matrix factorization, it is particularly difficult to measure or predict serendipity [Kotkov et al., 2016].

Fortunately, we can elegantly unite this definition with our causal framework and define it strictly as the value of an intervention. That is, we determine a recommendation to be serendipitous if it is an item that a user has a high true preference for (and will thus rate highly), *but for which the user has a very low probability of discovering without algorithmic intervention.* We can map these two requirements directly to the outputs of our causal model:

1. **Relevance (Likely to Like):** Represented by the unconfounded baseline preference, $\theta_u^\top \beta_i$. Because the exposure bias was absorbed by γ_u, this dot product represents the user's pure affinity for the item.
2. **Expectedness (Likely to Encounter):** Represented by the substitute confounder, $\hat{a}_{ui}$. This captures the probability that the user would naturally discover the item through mainstream popularity or existing habits (the confounding variables we provided examples for earlier).

From these requirements, we construct and propose a Serendipity Score (S_{ui}) for unseen items by maximizing unconfounded relevance while penalizing expectedness:

$$S_{ui} = \frac{\theta_u^\top \beta_i}{\max\{\hat{a}_{ui}, \epsilon\}}$$

where ϵ is a small constant that is dynamically set to the fifth percentile of all non-zero exposure probabilities in a given dataset. Here, a high S_{ui} indicates an item that the user is likely to like, yet unlikely to encounter. Conversely, a blockbuster that a user is nearly guaranteed to love but also has garnered a large volume of prerelease praise will yield a low S_{ui}, penalized by the large $\hat{a}_{ui}$.

Aside. As alluded to previously, we drop γ_u after training as a deliberate causal mechanism, effectively discarding exposure bias. During inference, we discard this term to isolate the pure, unconfounded preference $\theta_u^\top \beta_i$. Furthermore, while the existing literature has not formally defined a serendipity score via the ratio we present above, its form is supported by two strands of literature. First, it maps directly to a longstanding definition of serendipity: $Serendipity = Relevance \times Unexpectedness$ ([Ge et al., 2010]). We claim that unexpectedness is the inverse of expectedness in the context of how we have described it, so placing $\hat{a}_{ui}$ in the denominator satisfies this definition. Furthermore, the ratio we present mirrors Inverse Propensity Weighting (IPW) in the causal setting, where outcomes are weighed by the inverse of their own exposure propensity ($1/\hat{a}_{ui}$) to estimate causality ([Schnabel et al., 2016]). We apply Winsorization, or lower-bound capping, to the denominator so that if an item's exposure probability drops below our ϵ threshold, the item does not artificially inflate the serendipity score due to a near-zero denominator. Furthermore, we set ϵ to reflect the fifth percentile of all non-zero exposure probabilities observed in the dataset to ensure that our lower bound scales naturally with the inherent sparsity of a given dataset.

Recall from Section 2.2 that Johnson's approach resolves the practical limitations of our serendipity framework: rather than ignoring unobserved interactions ($r_{ui} = 0$) as traditional MF does, the LMF framework treats them as negative observations with low confidence. This acknowledges that a lack of interaction does not necessarily indicate a true negative preference, but rather a lack of awareness: the exact type of content that serendipity seeks to unearth. With the theoretical boundaries of causality defined, as well as the empirical tools for implicit feedback established via our modified logistic matrix factorization framework, we are ready to formally define our algorithm to construct a recommendation slate.

3 Algorithm

A combination of the theoretical frameworks outlined in Section 2 yields our proposed algorithm for a utility-constrained causal serendipity recommender system. Let U be the set of users and I be the set of items in our system. Let N denote the total number of recommendation slots available in the final user interface state.

3.1 Exposure Modeling ($\hat{a}_{ui}$)

We first process raw implicit feedback data to learn the true, unconfounded latent preferences of our users. Construct a binary exposure matrix A, where $a_{ui} = 1$ if a user has any historical interaction and 0 otherwise. Following Wang et al., we fit a Poisson factorization model to A to capture the underlying mechanisms of discovery. The output is $\hat{a}_{ui}$, or the expected probability that user u is naturally exposed to item i due to mainstream popularity or existing habits.

3.2 Confidence Formulation (c_{ui})

Drawing on Johnson's LMF framework, process the raw implicit interaction volume r_{ui} (e.g., total stream time) into a confidence metric:

$$c_{ui} = \alpha r_{ui}$$

where α is a predefined scaling hyperparameter.

3.3 Outcome Model Optimization

Initialize the latent user matrix Θ, latent item matrix B, and the user-specific confounder weights Γ. To isolate the unconfounded preference, inject our expected exposure probability $\hat{a}_{ui}$ from Step

3.1:

$$p(l_{ui}) = \sigma(\theta_u^\top \beta_i + \gamma_u \hat{a}_{ui})$$

To train these parameters, maximize the confidence-weighted log-likelihood $\mathcal{L}$ across all user-item pairs. Here, the confidence metric c_{ui} scales the importance of observed interactions against the sparse matrix:

$$\mathcal{L} = \sum_{u \in U} \sum_{i \in I} c_{ui} \Big(a_{ui} \log p(l_{ui}) + (1 - a_{ui}) \log(1 - p(l_{ui})) \Big)$$

As noted in the aside in Section 2.2, we optimize this via alternating gradient ascent. Exposure bias is absorbed into γ_u, and the unconfounded latent vectors θ_u and β_i are hereafter used for inference.

3.4 Causal Scoring and Sorting

For a target user u, isolate the set of unobserved items, defined as $I_{unobs} = \{i \in I \mid a_{ui} = 0\}$. For each item, calculate the serendipity score:

$$S_{ui} = \frac{\theta_u^\top \beta_i}{\max(\hat{a}_{ui}, \epsilon)}$$

where ϵ is a dynamic winsorization threshold set to the fifth percentile of all non-zero exposure probabilities observed in the dataset. This lower-bound capping prevents items with near-zero exposure from artificially inflating the serendipity score due to data sparsity.

3.5 Recommendation Slate Assembly

Initialize the final slate, denoted $Slate_u$, as an empty array. Sort the filtered candidate pool in descending order by their serendipity score S_{ui}. Select the top N items globally and append them to $Slate_u$.

4 Results

4.1 Dataset

We evaluate our algorithm defined in Section 3 on the Last.fm HetRec 2011 dataset, which contains social networking, tagging, and music artist listening information [Last.fm, nd, Cantador et al., 2011]. More specifically, the dataset includes 92,834 user-artist interactions across 1,892 users and 17,632 artists. Additionally, the raw listen count r_{ui} serves as our implicit feedback signal.

In the left panel of Figure 1, the distribution of r_{ui} is heavily right-skewed. Moreover, the median interaction is 260 listens, while the mean is much larger at 745 listens (Table 1), likely due to a small number of extremely heavy listeners. Despite this skew, the median of 260 listens is well above a one-time encounter, suggesting that most observed user-artist pairs reflect genuine, sustained engagement. Furthermore, the interaction matrix is extremely sparse at 99.72%, consistent with real-world implicit feedback datasets where users engage with only a small fraction of available items.

Notably, the right panel of Figure 1 reveals that virtually all 1,892 users have interacted with exactly 50 artists, suggesting a hard cap imposed during data collection. This uniformity in density of user activity is worth noting, as it constrains the variation available to the exposure model in Section 4.3.

Table 1: Descriptive Statistics for Artist Listening Counts (r_{ui})

Metric	**Value**
Count	92,834
Mean	745
Std. Deviation	3,751
Minimum	1
25% (Q1)	107
50% (Median)	260
75% (Q3)	614
Maximum	352,698

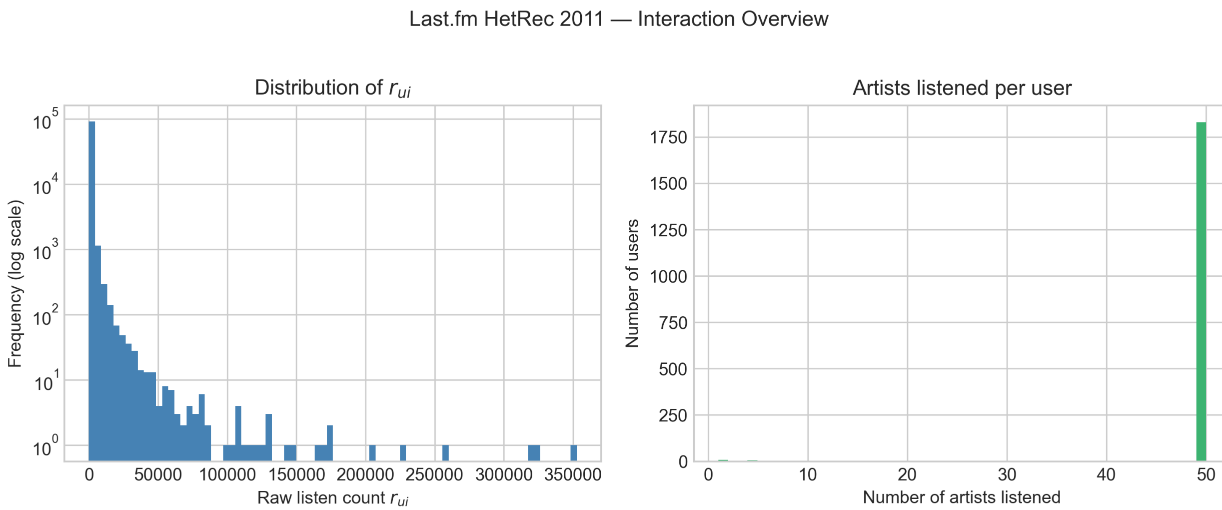

Figure 1: Last.fm HetRec 2011 Interaction Overview

4.2 Matrix Construction and Hyperparameter Decisions

Raw listen counts r_{ui} are scaled into confidence weights following Johnson's framework. Given the extreme right skew of the interaction distribution (Figure 1), we apply a log-scaling such that $c_{ui} = 1 + \alpha \log(1 + r_{ui})$, preventing the small number of very heavy listeners from dominating the gradient updates. We set $\alpha = 40$ following Johnson's original recommendation and train with $k = 20$ latent dimensions and 100 gradient ascent iterations.

4.3 Stage 1: Exposure Modeling

Hierarchical Poisson Factorization (HPF) is fit to the binary exposure matrix A with $k = 20$ latent dimensions. The resulting matrix $\hat{a}_{ui}$, after min-max normalization to $[0, 1]$, exhibits a distribution heavily concentrated near zero (mean $\hat{a}_{ui} = 0.0026$, median $\hat{a}_{ui} = 0.0008$) since most user-artist pairs have a very probability of low natural discovery, with only a small number of mainstream artists receiving consistently high exposure scores among users. This concentration is consistent with the long-tail structure of music consumption and validates that HPF captures the underlying popularity-driven mechanism.

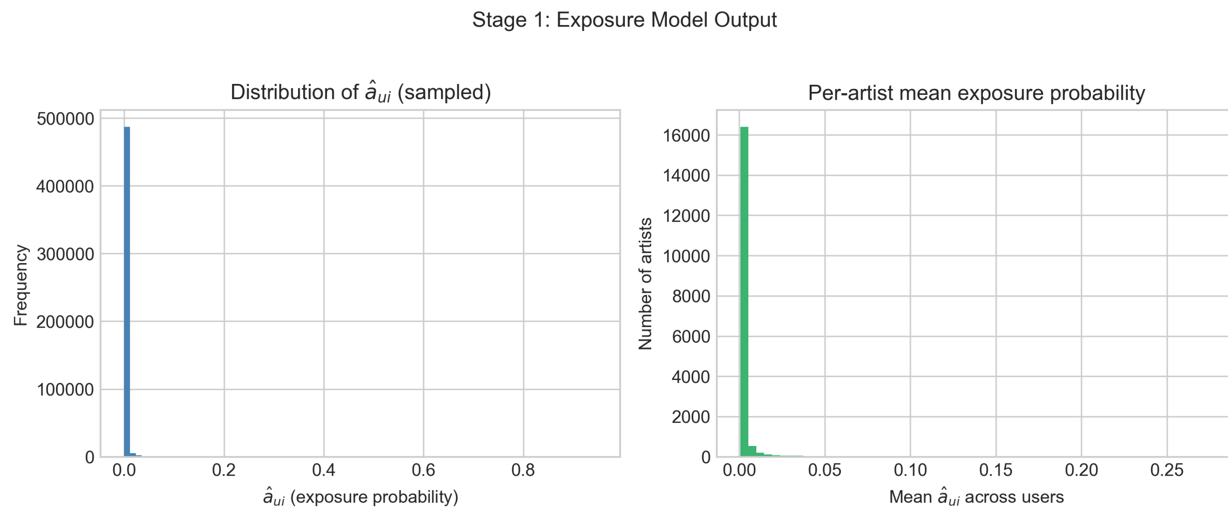

Figure 2: Stage 1: Exposure Model Output

The concentration of exposure probabilities in Figure 2 also motivates the winsorization analysis in Section 4.6.2, as items with near-zero $\hat{a}_{ui}$ risk producing artificially inflated serendipity scores.

4.4 Case Study: Target User 5

To test the effectiveness of the algorithm, we analyze User 5's profile and generate recommendations based on their interaction history, exemplified by their top 10 artists in Table 3. Additionally, User 5 is a predominantly pop-driven listener, as pop accounts for nearly half of their top 50 artists and an even higher percentage of the total plays shown in Table 2.

Table 2: Genre Breakdown for User 5's Top 50 Artists

Genre	Number of Artists	% of Total Plays
Pop	24	74.73
Other	18	15.89
Female Vocalists	4	4.21
Dance	2	2.75
Rock	1	2.42

Table 3: Top 10 Most-Listened Artists for User 5

Rank	Name	Play Count	Genre
1	Rihanna	43,864	pop
2	Britney Spears	11,112	pop
3	Jordin Sparks	6,266	pop
4	Lady Gaga	5,846	pop
5	Kelly Clarkson	5,379	pop
6	Mariah Carey	5,188	other
7	Christina Aguilera	4,201	pop
8	Ashlee Simpson	4,059	pop
9	Leona Lewis	3,251	female vocalists
10	Kylie Minogue	3,151	dance

4.5 Stage 2: Outcome Modeling

The Causal LMF is trained on the log-scaled confidence matrix C (with $\alpha = 40$) for 100 alternating gradient ascent iterations, warm-started from the HPF latent factors. The Standard LMF baseline is trained identically, but without the substitute confounder term $\hat{a}_{ui}$.

Comparing the two models in Tables 4 and 5, the Standard LMF ranks Maxwell and Bob Marley first and second with exposure probabilities of 0.75 and 0.80, respectively, suggesting that their high preference scores are partially driven by mainstream visibility. Moreover, the Causal LMF's top recommendation, Keane, carries a lower exposure probability of 0.02, indicating that absorbing $\hat{a}_{ui}$ into γ_u has successfully suppressed the popularity signal from the preference scores. Additionally, Céline Dion appears third in the Causal LMF slate with an exposure probability of 0.40. This edge case suggests that the causal model did not fully suppress her popularity despite relatively high mainstream visibility, implying that γ_u did not fully absorb her exposure signal, leaving some popularity bias in the preference score.

Table 4: Top 10 Recommendations: Standard LMF

Rank	Artist	Category	Pref. Score	Exposure Prob.
1	Maxwell	other	72.7736	0.7526
2	Bob Marley	other	33.5737	0.8028
3	James Blunt	pop	25.4255	0.0226
4	Amerie	other	24.5865	0.0012
5	50 Cent	other	24.2613	0.0008
6	Sandy Leah	other	21.8390	0.0011
7	Jessica Simpson	pop	21.2321	0.0005
8	Usher	other	20.5028	0.0046
9	John Legend	other	20.3304	0.0011
10	Taylor Swift	other	18.7732	0.0039

Table 5: Top 10 Recommendations: Causal LMF

Rank	Artist	Category	Pref. Score	Exposure Prob.
1	Keane	alternative	43.9542	0.0201
2	James Blunt	pop	31.1082	0.0226
3	Céline Dion	female vocalists	28.2859	0.3952
4	Ke$ha	pop	23.8611	0.0025
5	Shakira	pop	23.8044	0.0060
6	Michael Jackson	pop	23.7586	0.0018
7	Miley Cyrus	pop	21.6738	0.0006
8	Demi Lovato	pop	21.3733	0.0012
9	Coldplay	alternative	19.6955	0.0004
10	Linkin Park	rock	19.4635	0.0017

4.6 Stage 3: Serendipity Scoring and Slate Assembly

4.6.1 Slate Assembly

The final slates for Standard LMF, Causal LMF, and Causal LMF + Serendipity for User 5 consist of 10 artists, all unobserved. The slates predominantly recommend pop artists, though some—including Coldplay and Miley Cyrus—have a surprisingly low exposure probability given their mainstream prominence in 2011. For example, in Figure 3, Coldplay and Miley Cyrus appear as serendipitous recommendations for User 5, which is plausible given User 5's strong affinity for pop music. However, the degree to which their exposure probabilities are suppressed is likely an artifact of the limited scope of the dataset, as discussed in Section 5.2.

Table 6: Top 10 Recommendations: Causal LMF + Serendipity

Rank	Artist	Category	S_{ui}	Pref. Score	Exposure Prob.
1	Timbaland	other	51801.0859	18.5178	0.0004
2	Coldplay	alternative	49822.9375	19.6955	0.0004
3	James Morrison	other	40217.4336	14.9425	0.0004
4	Nickelback	rock	38618.1055	13.7679	0.0004
5	Miley Cyrus	pop	35284.1289	21.6738	0.0006
6	Duffy	other	33704.1484	13.3236	0.0004
7	Glee Cast	other	30643.9805	15.8417	0.0005
8	Depeche Mode	electronic	29392.4590	11.6191	0.0004
9	Björk	electronic	28987.9336	10.9605	0.0004
10	The Killers	indie	28328.0547	11.4101	0.0004

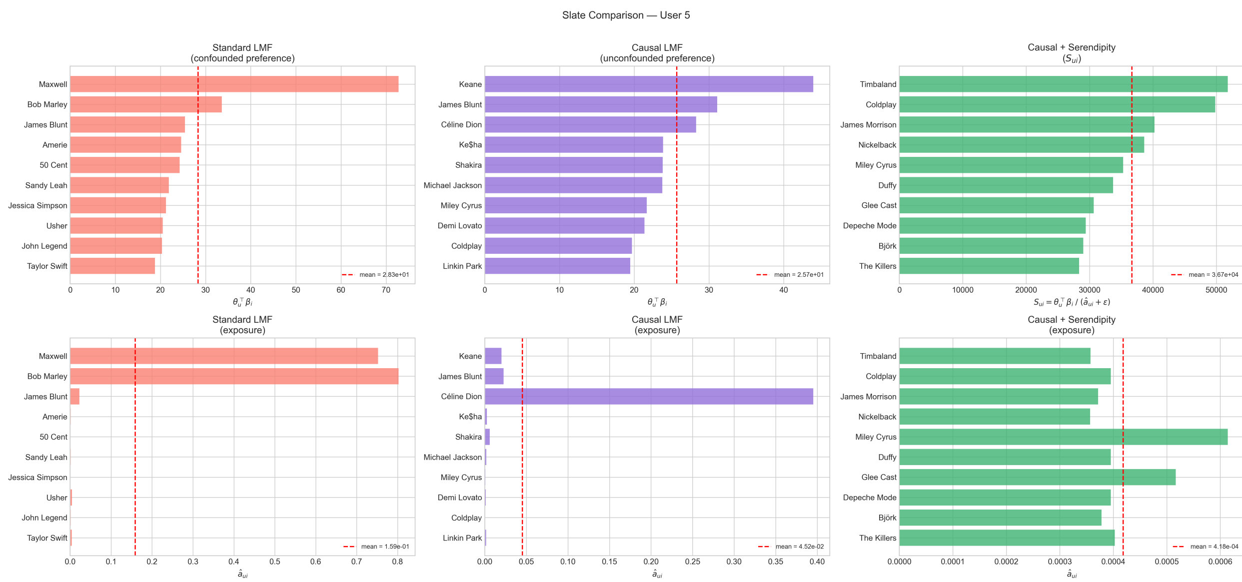

Figure 3: Slate Comparison for User 5

4.6.2 Winsorization

Applying S_{ui} without a denominator floor (i.e., $S_{ui} = (\theta_u^\top \beta_i)/(\hat{a}_{ui} + \epsilon)$) produces extreme outliers for artists with near-zero exposure estimates. Using an additive-shift formulation with $\epsilon = 10^{-8}$ allows scores to reach astronomically high values for items with $\hat{a}_{ui}$ in the range $[10^{-9}, 10^{-5}]$, regardless of preference quality. When this occurs, the ranking is effectively determined by the exposure denominator ($\hat{a}_{ui}$) rather than the preference numerator ($\theta_u^\top \beta_i$), undermining the score's intent to surface genuinely preferred but undiscoverable items.

Thus, replacing the additive shift with a hard winsorization floor—either the fixed constant $\epsilon = 0.01$ or the data-driven fifth percentile of non-zero exposures—stabilizes the score distribution while preserving the rank ordering among genuinely low-exposure candidates.

Compared to the baseline serendipity slates (Table 6), the scores produced by the dynamic floor (Table 8) remain very similar in both magnitude and recommendation order. This occurs because the dynamic floor of $\epsilon = 0.000326$ is already close to the minimum observed exposure values in the dataset, effectively leaving the ranking unchanged.

Conversely, the fixed floor of 0.01 (Table 7) is high enough to aggressively clip the exposure denominators of several low-exposure artists. This drastically reduces the overall magnitude of the S_{ui} scores and elevates artists with moderately higher exposure probabilities—such as Ke$ha, Shakira, and Michael Jackson—who are more recognizable, and therefore less serendipitous for a pop-oriented listener like User 5.

Table 7: Top 10 Recommendations: Causal LMF + Serendipity (Fixed Floor: 0.01)

Rank	Artist	Category	S_{ui}	Pref. Score	Exposure Prob.
1	Ke$ha	pop	2386.11	23.8611	0.002455
2	Shakira	pop	2380.44	23.8044	0.006012
3	Michael Jackson	pop	2375.86	23.7586	0.001838
4	Keane	alternative	2185.01	43.9542	0.020116
5	Miley Cyrus	pop	2167.38	21.6738	0.000614
6	Demi Lovato	pop	2137.33	21.3733	0.001244
7	Coldplay	alternative	1969.55	19.6955	0.000395
8	Linkin Park	rock	1946.35	19.4635	0.001674
9	Taylor Swift	other	1895.63	18.9563	0.003939
10	Timbaland	other	1851.78	18.5178	0.000357

Table 8: Top 10 Recommendations: Causal LMF + Serendipity (Dynamic Floor: 0.000326)

Rank	Artist	Category	S_{ui}	Pref. Score	Exposure Prob.
1	Timbaland	other	51802.54	18.5178	0.000357
2	Coldplay	alternative	49824.20	19.6955	0.000395
3	James Morrison	other	40218.52	14.9425	0.000372
4	Nickelback	rock	38619.19	13.7679	0.000357
5	Miley Cyrus	pop	35284.70	21.6738	0.000614
6	Duffy	other	33705.00	13.3236	0.000395
7	Glee Cast	other	30644.57	15.8417	0.000517
8	Depeche Mode	electronic	29393.20	11.6191	0.000395
9	Björk	electronic	28988.70	10.9605	0.000378
10	The Killers	indie	28328.76	11.4101	0.000403

In Table 9, Coldplay's S_{ui} collapses from 5,188.0 to 570.5 under the fixed floor, confirming that its score is sensitive to the choice of floor and partially driven by a near-zero exposure denominator. In contrast, the dynamic floor leaves Coldplay's score nearly unchanged at 5,188.1, since the fifth percentile floor of 0.000326 falls below Coldplay's already low exposure probability. Similarly, Björk's score drops from 3,289.6 to 291.8 under the fixed floor, while remaining stable under the dynamic floor. On the other hand, Rihanna's score is completely unchanged across all three scenarios at 3,963.3, as her exposure probability is high enough that neither floor clips her denominator. Thus, this illustrates that the score correctly penalizes discoverable items even when preference is high. Overall, artists with low exposure probability and sufficiently high preference scores yield the highest serendipity scores, while those with low preference are not regarded as serendipitous regardless of their exposure level.

Table 9: Spotlight Artist S_{ui} Across Winsorization Regimes

Artist	Role	Original	Fixed Floor	Dynamic Floor
Coldplay	Anomaly	5,188.0	570.5	5,188.1
Rihanna	Crowd Favorite	3,963.3	3,963.3	3,963.3
Björk	Hidden Gem	3,289.6	291.8	3,289.6

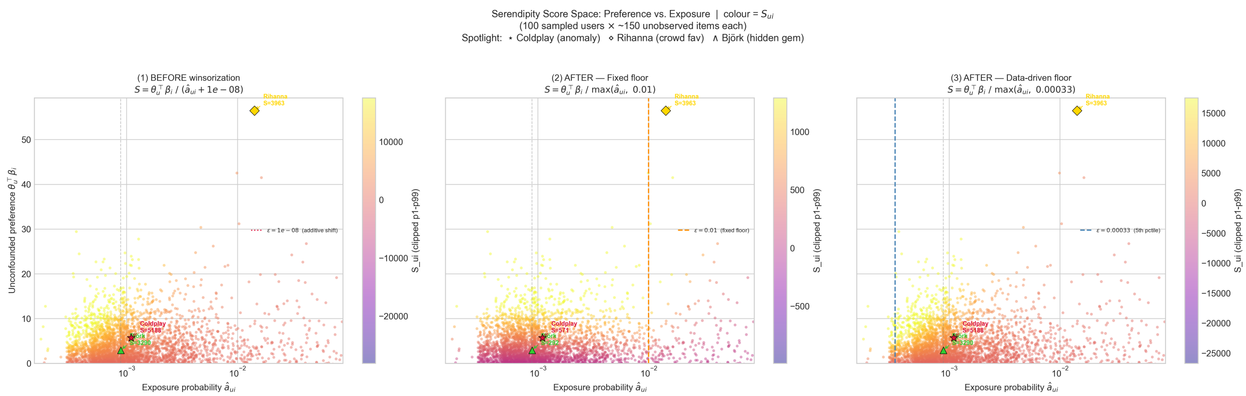

Figure 4: Serendipity Score Space: Preference vs. Exposure Across Winsorization Scenarios

These three plots together (Figure 4) illustrate that winsorization is a parameter choice with a meaningful trade-off. Ultimately, a tighter floor produces more conservative, preference-dominated slates, while a looser floor preserves the full unexpectedness signal at the cost of occasionally surfacing items whose high scores are driven more by near-zero exposure estimates than by genuine preference. Hence, we recommend the dynamic floor as the preferred formulation, as it adapts to the dataset and does not require manual tuning of ϵ, making it more generalizable between datasets with varying sparsity profiles.

4.7 Discussion

The results of our case study demonstrate illustrate the challenges of engineering serendipity in implicit recommender systems. Primarily, the contrast between the Standard LMF and Causal LMF slates (Tables 4 and 5) validates Wang et al.'s framework: by explicitly modeling the exposure mechanism, the algorithm suppressed the overarching popularity signal, isolating a purer representation of the user's latent preferences. The shift from cultural mainstays to highly relevant, but less exposed artists suggests that popularity bias can be decoupled from affinity.

However, the application of our proposed Serendipity Score (S_{ui}) reveals a sensitivity to data sparsity. As illustrated by the "Coldplay Anomaly" in Table 9, inverse-propensity weighting is unstable when applied to the long-tail distributions typical of implicit feedback datasets. Without lower-bound capping, unexpectedness can overpower unconfounded preference. Our winsorization analysis finds that while an additive shift leaves the system vulnerable to extreme outliers, a dynamic floor stabilizes the metric.

As a final note, while our dynamically winsorized serendipity score surfaces hidden gems (e.g., Björk and Depeche Mode for a pop-oriented listener), deploying such a slate in a live production environment risks inducing user fatigue. A slate composed entirely of unexpected items, no matter how relevant, violates the user's baseline consumption constraints. This reality necessitates the structural enhancements proposed in Section 5, ensuring that our causally derived serendipity is safely packaged within a utility-aware interface.

5 Further Enhancements & Limitations

5.1 The Utility of Utility Constraints

Note that the algorithm we share assembles the recommendation slate in order of our proposed serendipity score. While this allows us to propose unexpected, highly relevant items, simply ranking recommendations by S_{ui} in a descending order will likely not serve as an effective recommender system: we risk presenting a slate of obscure content to the user, who may be disincentivized to select anything due to unfamiliarity. To solve for this challenge, we consider future incorporation of the concept of utility.

Peng et al.'s consumption constraint framework perfectly complements the issues that arise from serving the top-N recommendations based on serendipity. Because users typically only consume one item in a session (e.g., one movie a night, one song at a time) they define true utility, denoted as $util_1$, not as the average accuracy of the algorithm's list of recommendations, but as the probability that the user likes at least one item in the recommended set. Grouping candidate items into m categories, where category t has conditional user satisfaction rate q_t, and n_t represents the number of slots allocated to that category, the utility of the slate is defined as the complement of the user disliking everything [Peng et al., 2024]:

$$util_1 = 1 - \prod_{t=1}^{m} (1 - q_t)^{n_t}$$

Peng et al. propose what they refer to as their "Milk and Ice Cream" corollary: to maximize $util_1$, the recommender system must allocate its N recommendation slots disproportionally to a category's satisfaction rate. That is, if a category has a low q_t, it requires a higher n_t (more slots) to ensure the user's consumption constraint is met [Peng et al., 2024].

In the context of serendipity, categories that yield high serendipity scores (e.g., obscure foreign films) naturally have a lower conditional satisfaction rate q_t than mainstream categories. While the user may enjoy *Your Name Engraved Herein*, they may not more broadly enjoy martial law-era Taiwanese films.

By utilizing Johnson's confidence weighting (c_{ui}), we can objectively define what constitutes a satisfying interaction in an implicit setting. Rather than trying to predict Peng's category satisfaction rate (q_t), we can empirically calculate it by defining a threshold of confidence (e.g., streaming a song at least 75% of the way through). q_t then becomes formalized as the historical conversion rate of exposures to high-confidence interactions within the category.

Thus, we aim to generate the final slate of recommendations via utility-constrained serendipity optimization. This can be incorporated into our algorithm as follows:

5.1.1 Empirical Estimation of Category Quality (q_t)

Establish the baseline reliability of each item category $t \in \{1, 2, \ldots, m\}$. Utilizing the implicit confidence formulation, calculate q_t globally as the historical conversion rate of exposures to high-confidence interactions. For a given category t:

$$q_t = \frac{\sum_{u \in U} \sum_{i \in t} \mathbb{I}(c_{ui} \geq \tau_c)}{\sum_{u \in U} \sum_{i \in t} a_{ui}}$$

where τ_c is a predefined hyperparameter threshold representing a "satisfying" interaction and c_{ui} is the user's implicit interaction confidence for the item (derived in Step 3.2).

5.1.2 Proportional Slot Allocation (n_t)

To maximize the utility of our slate of recommendations ($util_1$) while ensuring the inclusion of serendipitous categories, operationalize Peng et al.'s corollary via an inverse-weighting function. Calculate the raw inverse weight w_t for each target category:

$$w_t = \frac{1}{q_t}$$

Next, normalize weights to distribute exactly N integer slots across the available categories:

$$n_t = \text{round}\left(N \times \frac{w_t}{\sum_{k=1}^{m} w_k}\right)$$

To handle rounding issues where $\sum n_t \neq N$, add or subtract the integer remainder from the category with the highest serendipity variance until exactly N slots are allocated.

5.1.3 Compose Recommendation Slate: Constrained Sorting

Rather than sorting the candidate items globally as outlined in the base algorithm, we modify the slate assembly to enforce our new utility quotas. Group the eligible, filtered unobserved items by their respective categories t, and sort each group independently in descending order based on their S_{ui} score. Initialize the final slate $Slate_u$. Iterate through each category t, select the top n_t items from that category's sorted list, and append them to $Slate_u$ until exactly N slots are filled.

5.2 Limitations

Several limitations of the work done in Section 4 are worth acknowledging. First, the Last.fm HetRec 2011 dataset appears to limit users to 50 artist interactions, as shown by the right panel of Figure 1. This uniformity artificially constrains the variation available to the exposure model, as HPF relies on differences in user activity density to learn meaningful latent structure. In a less constrained dataset, users with broader or narrower listening histories would provide stronger signals to differentiate popularity-driven discovery from genuine preference.

Additionally, while we follow Johnson's original recommendation of $\alpha = 40$ for the confidence scaling hyperparameter, this value was not tuned to the specific distributional properties of the Last.fm dataset. Similarly, the fixed winsorization floor of $\epsilon = 0.01$ and the choice of the fifth percentile for the dynamic floor were not optimized. Ultimately, experimenting with higher fixed floors or alternative percentile thresholds could meaningfully shift the composition of the serendipity slate.

Finally, all results are demonstrated on a single user, User 5, which limits the generalizability of our findings. Without evaluating across a representative sample of users, it is difficult to determine whether the serendipity mechanism performs consistently or whether the results are unique to this particular user's listening history. Further evaluation would require testing across many users and ideally validating against held-out interactions or real user feedback. This connects to a deeper limitation of the framework, as there is no ground truth for serendipity. The score is theoretically supported, but whether serendipitous recommendations are actually experienced as surprising and satisfying by real users remains unverified. Thus, a user study would be necessary to confirm this.

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